

EXPERIENCE

Indian Space Research Organization

Ahmedabad, Gujarat

AI Research Intern

February 2026 - August 2026

- Consolidated a multi-stage atmospheric stability index calculation into a single step AI predictive task, achieving an average R2 score of 94.78%, by developing a custom neural network for the direct retrieval of the stability indices using satellite observations.
- Optimized model predictive reliability on unlabeled satellite observed data, reducing the RMSE from 8.6% to 5% by implementing rigorous five-fold cross validation, and hyperparameter tuning across the ensemble architecture.

Archsoft

Raipur, Chhattisgarh

AI and Machine Learning Intern

June 2025 - September 2025

- Shipped a full stack AI forensics application (async file handling, SQLite persistence, and FastAPI), with **sub-second inference** and real-time GradCAM explainability on a 22.2M parameter model.
- Achieved 93.85% ROC-AUC score in detecting AI-generated images by engineering a dual-stream architecture fusing EfficientNet-B4 spatial features with mid-frequency DCT-coefficients, and made the accuracy/latency trade-off explicit for production use.

FREELANCE PROJECTS

Multi-Process Validation Harness for Distributed Training — (Python, PyTorch, DeepSpeed, Tensorboard)

- Designed a Harbor benchmark to model FSDP checkpoint-resume state desynchronizations in EMA pipelines, validating synthetic self-distillation and reference fixes, measured by harbor agent trials scoring 1.0 and 0.0, **to evaluate agent performances**.
- Developed a multi-process **PyTorch distributed harness for deterministic training** and checkpoint recovery, guaranteeing exact per-rank state synchronization to a strict $1e-4$ numerical tolerance.

PROJECTS

Transformer Training and Gradient Optimization Infrastructure — (Python, PyTorch, Transformers)

- Eliminated silent loss drift in mixed-precision transformer training by refactoring **PyTorch GradientRelay wrappers**, restoring loss trajectory fidelity to less than a 2% relative error margin across 1, 4, and 8 micro-batch configurations.
- Stabilized **Automatic Mixed Precision (AMP)** scaler hooks during multi-step backward passes, preventing an 87.5% gradient magnitude suppression across 8-span accumulation steps.
- Engineered a deterministic verification framework for mixed-precision convergence, validating execution traces and gradient norms across distinct checkpoints.

Containerized Security Evaluation and Threat Modeling — (Python, Docker, Cryptography, Harbor-testing)

- Fine-tuned elliptic curve cryptanalysis payloads by implementing LLL lattice reduction and Babai's Nearest Plane Algorithm to solve the Closest Vector Problem.
- Automated the deterministic recovery of 256-bit private keys across the security pipeline, overcoming sparse datasets of only 55 signatures with 5 leaked most-significant bits in polynomial sub-second execution time.
- Engineered a containerized benchmarking environment for autonomous AI models, **orchestrating multi-stage agent trials** for rigorous cryptographic evaluation under strict runtime budgets.

DataPrepX: AI-Driven Data Automation Platform — (Python, FastAPI, TensorFlow, Streamlit)

- Built a data preparation automation platform to **streamline predictive modeling workflows**, maintaining 86.74% processing accuracy and reducing manual effort by 70% across diverse datasets.
- Designed a **data-aware model** selection engine orchestrating 7 adaptive regression and classification models leveraging **LLM-Driven Summarization** and **Model Explainability**.
- Implemented a comprehensive software validation framework, to systematically test core processing, and estimations.

TECHNICAL SKILLS

- Languages and Tools:** Python, GoLang, SQL, LangChain, LangGraph, FastAPI, PyTorch, TensorFlow, Docker, Kubernetes, Git/GitHub, Flask, React, Harbor-testing
- Databases:** SQLite, Supabase, MongoDB, MySQL, PostgreSQL
- Operating Systems:** MacOS(Unix), Red Hat-8, Linux

EDUCATION

Vishwakarma Institute of Technology

Pune, Maharashtra

B.Tech — Information Technology (CGPA - 8.29)

October 2022 - June 2026

PUBLICATIONS

Published paper (ICAEET-NCCT-2025): **AI Based Voice Interviewer for Response Evaluation**

Published paper (IEEE ICESCI 2025): **Securing Automotive Using Iris Recognition**